



Module 4 – Rafay Fundamentals

Enterprise Kubernetes Operations Platform

DEVOPS · PLATFORM · CLOUD ENGINEERING



Learning Objectives

01

Why Rafay?

Understand the enterprise challenges that Rafay solves across multi-cluster Kubernetes environments.

02

Architecture

Learn the Controller, Agent, and how secure communication flows between components.

03

Blueprints & Templates

Standardize cluster configuration and application deployments across your fleet.

04

Cluster Registration

Onboard existing Kubernetes clusters into Rafay for centralized enterprise management.

The Multi-Cluster Challenge

Managing one cluster is easy. Managing hundreds requires automation.

Without a Platform

- Multiple kubeconfigs to maintain
- Different cloud providers with inconsistent tooling
- Manual RBAC and security policies per cluster
- No standardization or governance
- Operational complexity at scale

The Enterprise Reality

Organizations running Kubernetes at scale face fragmented visibility, inconsistent security posture, and significant operational overhead. Each cluster becomes a snowflake — unique, hard to manage, and difficult to audit.



Operational complexity grows exponentially with each additional cluster added to the fleet.

What is Rafay?

Rafay is an **Enterprise Kubernetes Operations Platform** that provides a single control plane for managing Kubernetes clusters across any infrastructure.



Amazon EKS

Managed Kubernetes on AWS, including EKS Anywhere for on-premises.



Azure AKS

Microsoft Azure managed Kubernetes service.



Google GKE

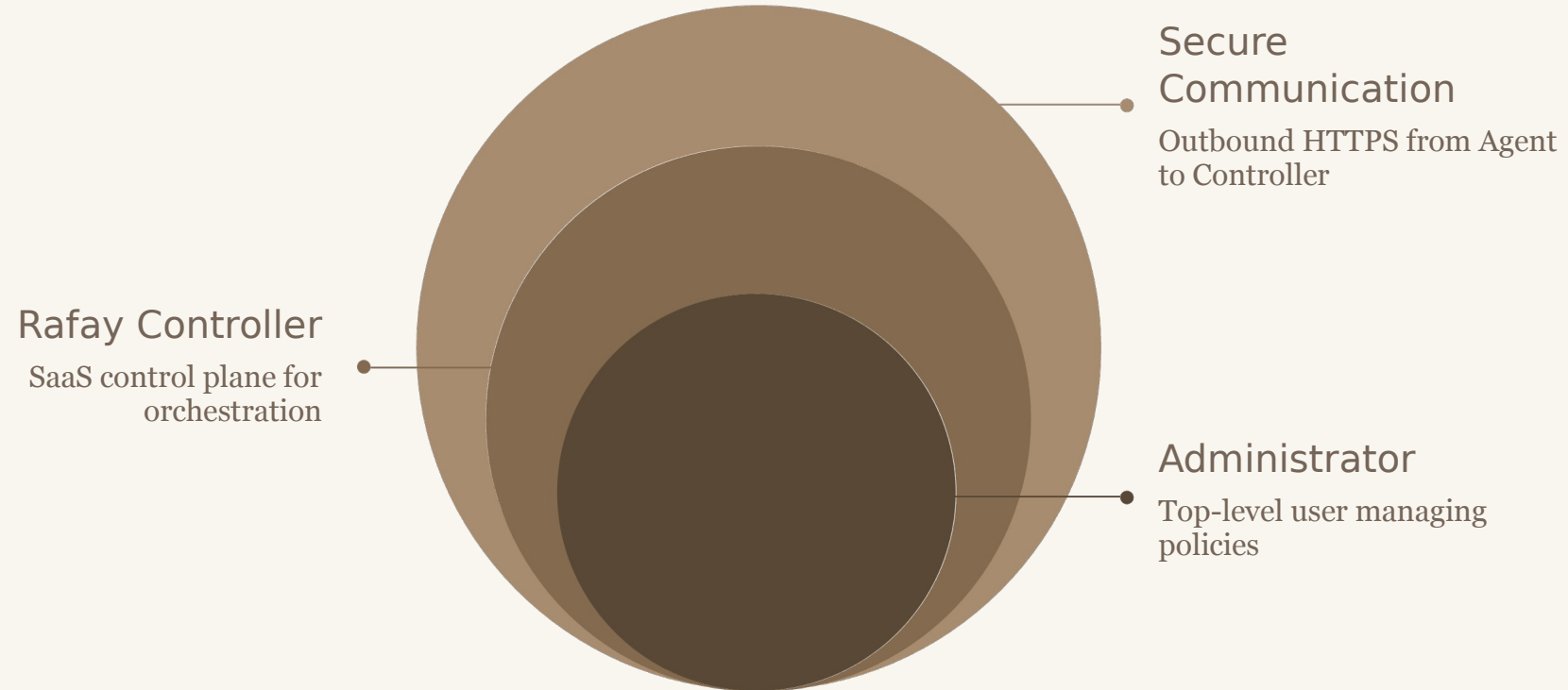
Google Cloud managed Kubernetes engine.



OpenShift & Edge

Red Hat OpenShift, upstream Kubernetes, and edge deployments.

Rafay Architecture Overview



All communication is **outbound from the Agent to the Controller** — no inbound firewall ports are required. This architecture enables secure connectivity even in locked-down enterprise and edge environments.

- ❑ No inbound firewall ports required. The Agent initiates all connections to the Controller over HTTPS.



Rafay Core Components

Controller

The SaaS brain of the platform.
Manages authentication, authorization, policies, blueprints, templates, and cluster inventory.

Agent

Installed inside each cluster.
Executes deployments, reports health, syncs configuration, and maintains secure communication with the Controller.

⚙️ Managed Cluster

Any Kubernetes cluster registered with Rafay — EKS, AKS, GKE, EKS Anywhere, OpenShift, or upstream K8s.

Rafay Controller Responsibilities

Platform Management

- Authentication & Authorization
- Role-Based Access Control (RBAC)
- Projects & Policies
- Audit Logs

Deployment Management

- GitOps workflows
- Blueprints & Templates
- Cluster Inventory
- Deployment orchestration



- ❏ The Controller is delivered as a SaaS service — no infrastructure to manage for the control plane itself.

Rafay Agent

The Rafay Agent is a lightweight component **installed inside each Kubernetes cluster**. It acts as the secure bridge between the cluster and the Controller.

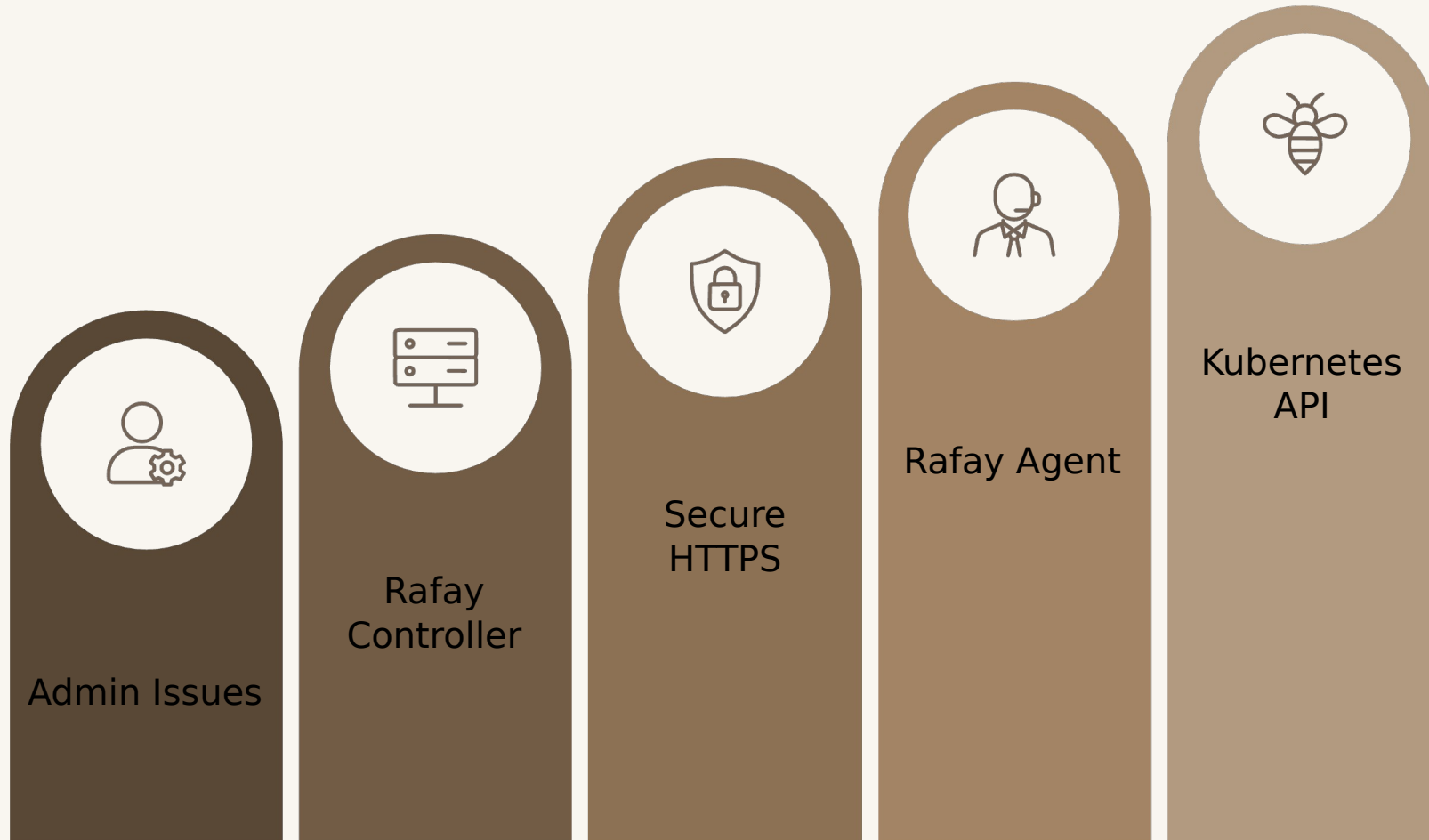
Agent Responsibilities

- Initiates secure HTTPS connection to Controller
- Executes deployments and configuration changes
- Reports cluster health and status
- Sends logs and events upstream
- Syncs Blueprint and Template configurations

📄 The Agent initiates communication with the Controller — never the other way around.



Communication Workflow



Every action flows through this secure chain. The Administrator never communicates directly with the cluster — all operations are mediated by the Controller and executed by the Agent. This ensures consistent policy enforcement, auditability, and security across every cluster in the fleet.

Blueprints

A **Blueprint** is a reusable cluster configuration template that defines the standard platform services applied to every cluster in your organization.

Access & Security

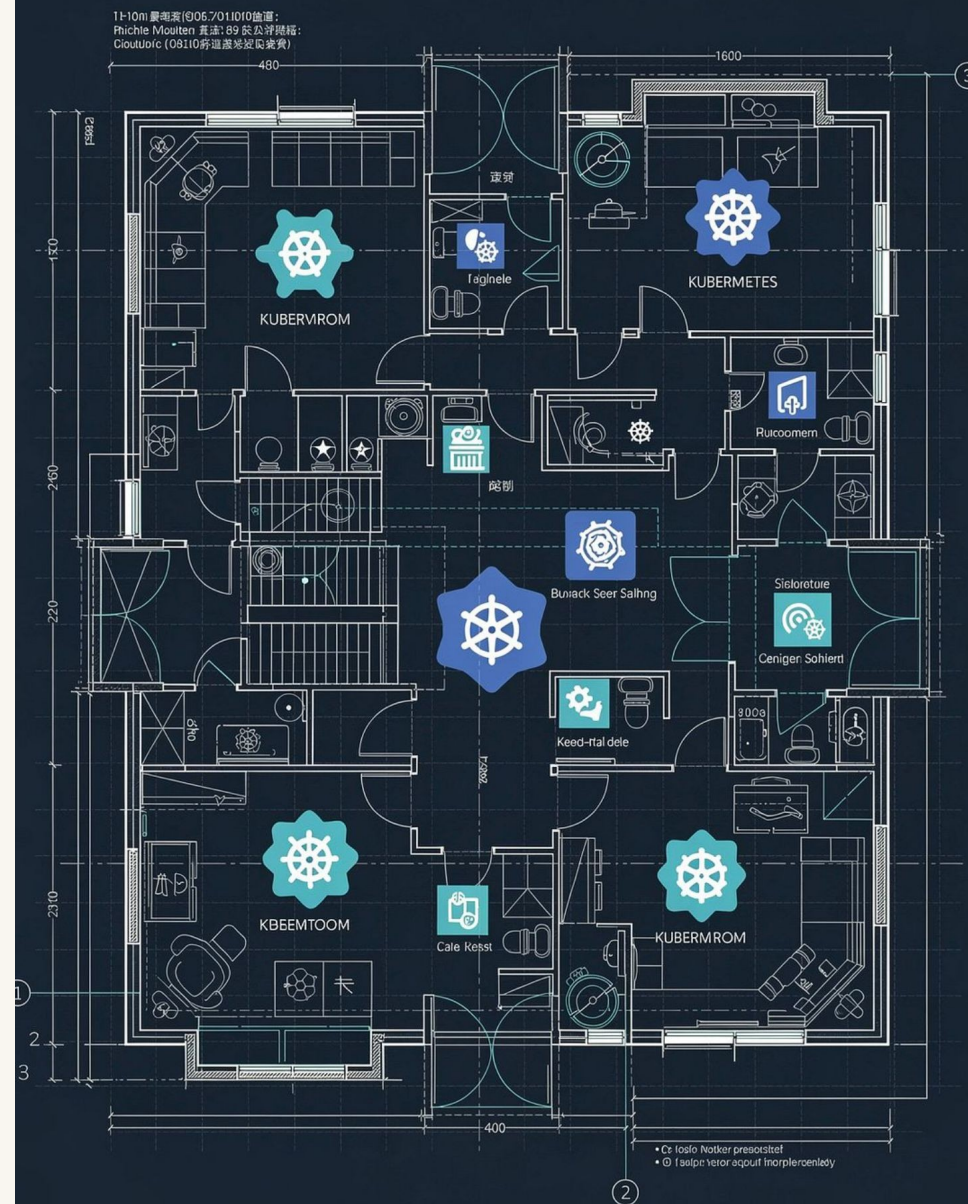
- Namespaces & RBAC
- OPA Policies
- Network Policies

Storage & Networking

- Storage Classes
- Ingress Controllers

Observability

- Monitoring stack
- Logging agents



Benefits of Blueprints

WITHOUT BLUEPRINTS

Manual Configuration

- Each cluster configured individually
- Inconsistent security policies
- High risk of human error
- Slow onboarding of new clusters
- Difficult to audit and govern

1

Blueprint

Applied consistently across hundreds of clusters

WITH BLUEPRINTS

Reusable Standardized Platform

- One Blueprint applied to all clusters
- Consistent compliance posture
- Reduced configuration errors
- Faster cluster onboarding
- Full governance and auditability

100%

Consistency

Every cluster meets the same platform standard



Reduced Errors

Eliminate manual per-cluster configuration drift

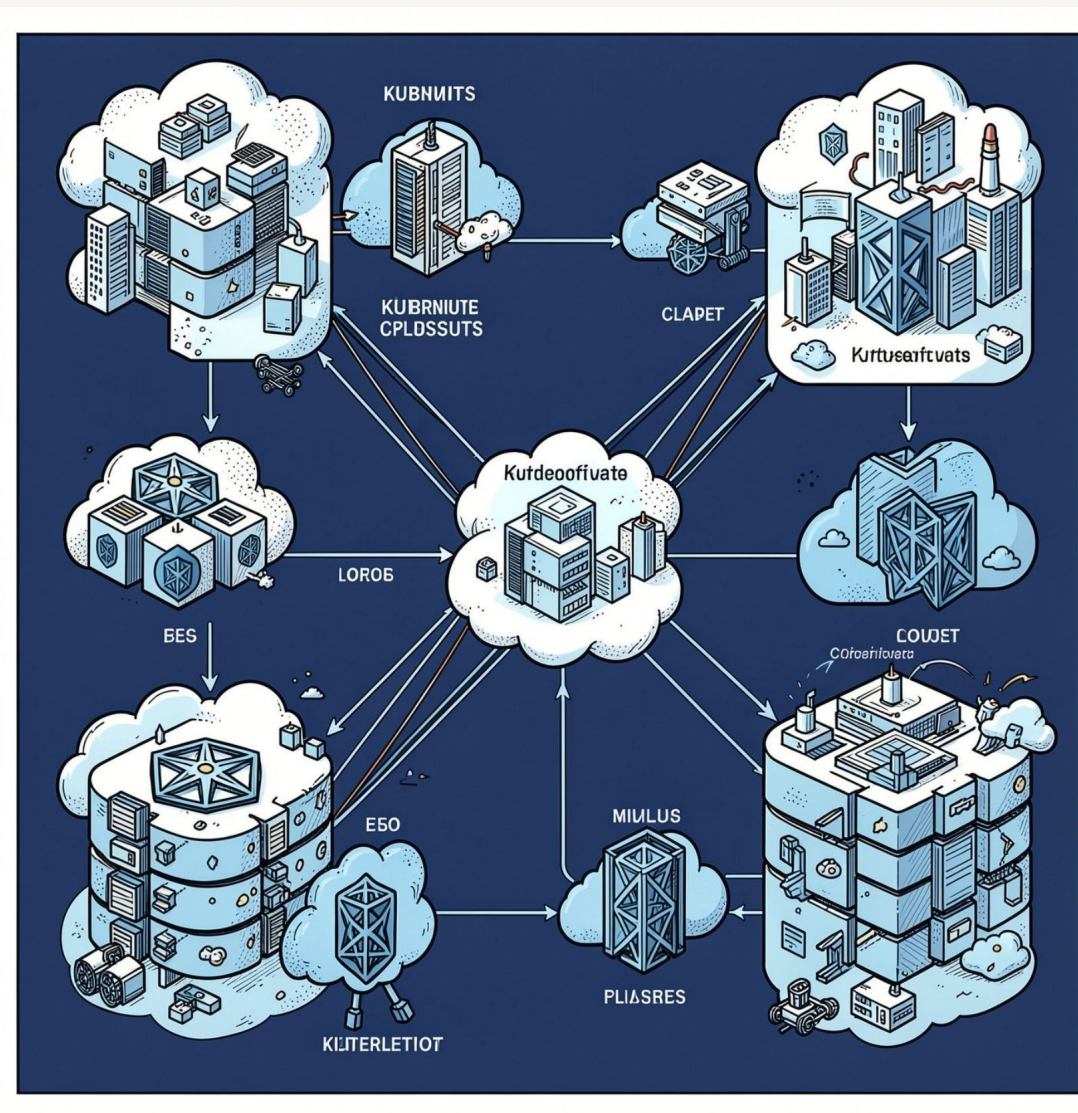
Templates

Templates define reusable application deployment configurations that can be deployed consistently across multiple clusters.

Template Components

- Helm Charts
- Kubernetes YAML manifests
- Variables & Parameters
- ConfigMaps & Secrets
- Services & Deployments

❏ One Template definition can deploy the same application to dozens of clusters with environment-specific variable overrides.



Blueprint vs. Template

Dimension	Blueprint	Template
Layer	Platform layer	Application layer
Purpose	Cluster configuration	Workload deployment
Scope	Infrastructure services	Reusable app configuration
Shared by	Shared platform team	Application teams

Analogy: Blueprint = Building Design | Template = Furniture Arrangement

Cluster Registration Workflow



Cluster registration is the process of onboarding any existing Kubernetes cluster into Rafay. The workflow is straightforward: a registration manifest is generated by the Controller and applied to the target cluster, which installs the Agent and establishes the secure connection automatically.



Enterprise Use Case

A global enterprise manages clusters across **AWS EKS, Azure AKS, Google GKE, EKS Anywhere, and Edge Kubernetes** — all through a single Rafay control plane.



Centralized Visibility

Unified dashboard across all clusters and clouds.



Security & Governance

Consistent RBAC, policies, and audit logs everywhere.



GitOps

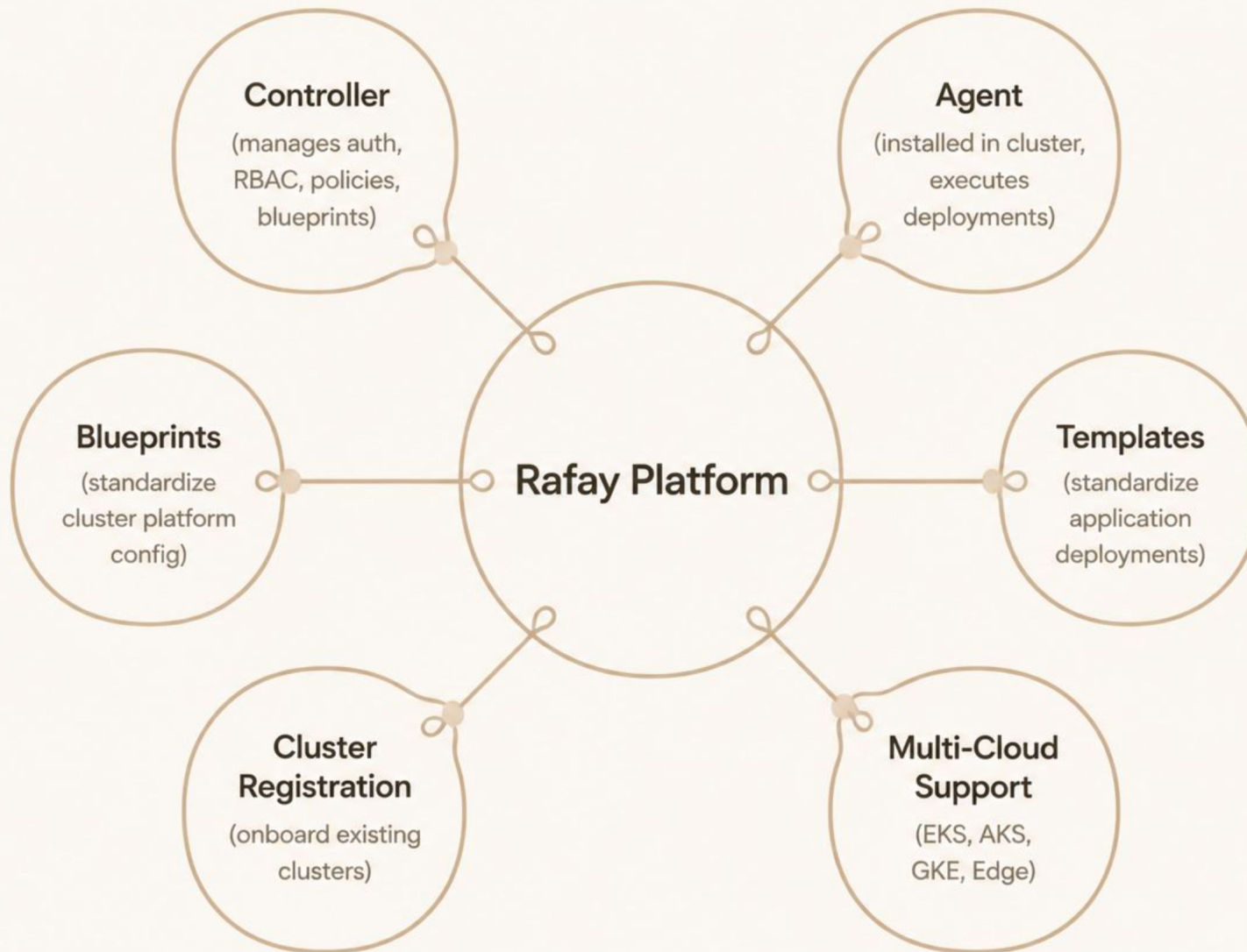
Declarative, version-controlled deployments at scale.



Standardized Deployments

Blueprints and Templates ensure consistency fleet-wide.

Key Concepts Recap



Architecture Best Practices

Security First

Always rely on the Agent's outbound-only communication model.
Never open inbound firewall ports to your clusters for Rafay connectivity.

Blueprint Governance

Define Blueprints at the platform team level. Treat them as code — version-control them in Git and review changes through a pull request workflow.

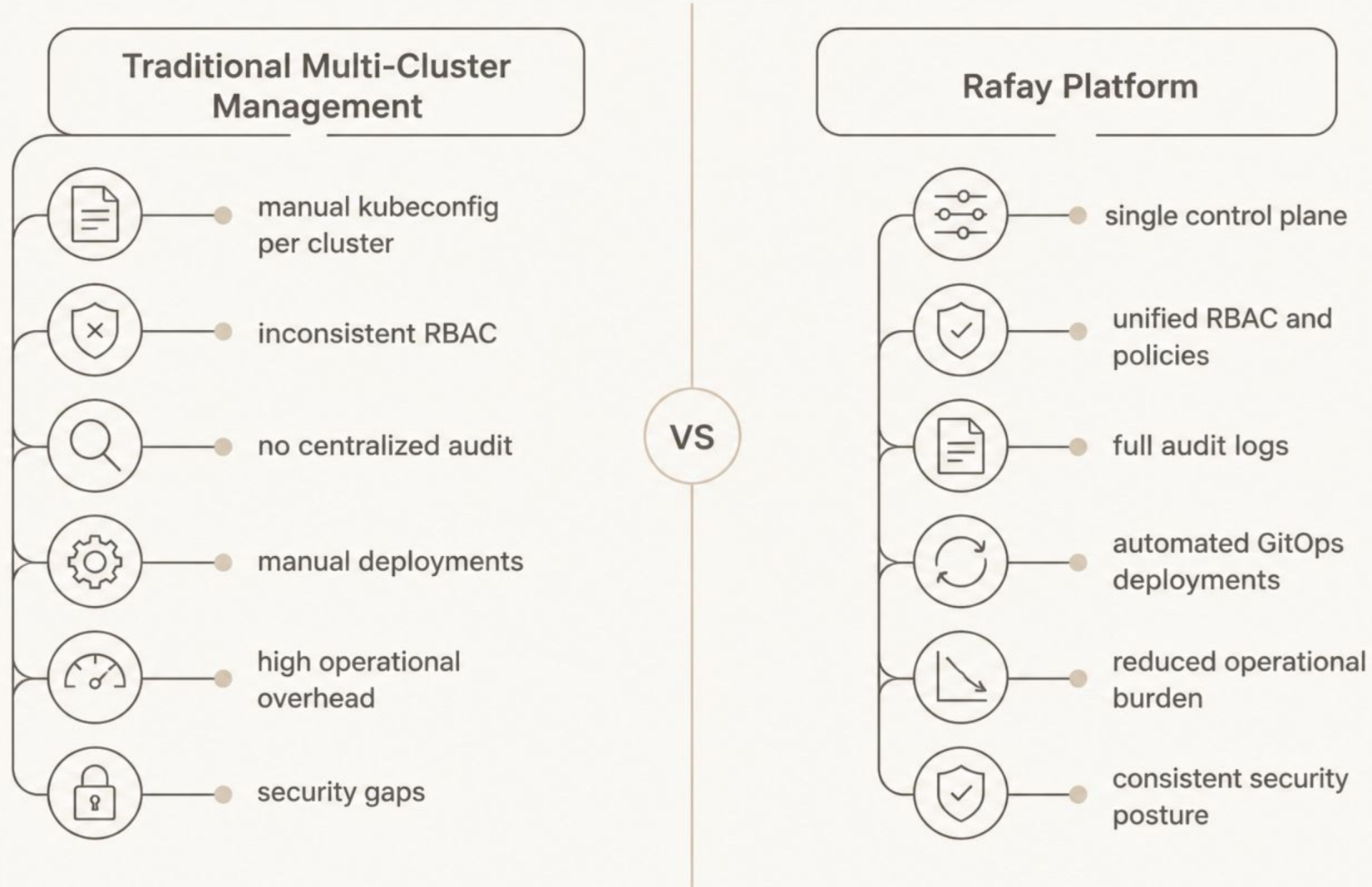
Template Parameterization

Use variables and parameters in Templates to support environment-specific overrides (dev, staging, prod) without duplicating Template definitions.

Project Isolation

Use Rafay Projects to enforce team and workload isolation, applying RBAC boundaries that mirror your organizational structure.

Rafay vs. Traditional Approaches



Rafay eliminates the operational overhead of managing Kubernetes clusters individually, replacing fragmented tooling with a single, governed enterprise platform.

Module 4 – Summary

→ Centralized Operations

Rafay provides a single control plane for all Kubernetes clusters across any cloud or infrastructure.

→ Blueprints & Templates

Blueprints standardize cluster platform configuration; Templates standardize application deployments across the fleet.

→ Controller & Agent

The Controller manages the platform; the Agent securely connects each cluster with outbound-only HTTPS communication.

→ Cluster Registration

Existing Kubernetes clusters — EKS, AKS, GKE, EKS Anywhere, Edge — can be onboarded via a simple registration manifest.



What's Next

Next Module

Hands-on Cluster Registration and Blueprint Deployment

In the next module, you will register a real Kubernetes cluster with Rafay, apply a Blueprint, and deploy an application using a Template — putting everything covered in Module 4 into practice.

Lab 1

Register an EKS
Anywhere cluster with
Rafay

Lab 2

Create and apply a
Blueprint to your
cluster

Lab 3

Deploy an application
using a Rafay
Template